VNR VJIET
VALLURUPALLI NAGESWARA RAO VIGNANA JYOTHI INSTITUTE OF ENGINEERING & TECHNOLOGY

CPDT
Centre for Presencing & Design Thinking

DESIGN THINKING ENGINEERING PROJECT · 2026

Find My Professor · VNR Connect

Dynamic QR Office Placards & Real-Time Consultation Scheduling Ecosystem for Modern Engineering Campuses

STUDENT ENGINEERING TEAM (CSE & AI)

Sai Koushik Pallapolu | Team Member 2 | Team Member 3 | Team Member 4

UNDER THE GUIDANCE OF

Mrs. Prasanna Pabba
Assistant Professor, Dept. of CSE · VNR VJIET

The Team

Centre for Presencing and Design Thinking (CPDT) · VNR VJIET



Our Innovation Commitment

Driven by real campus pain points, our team applied the human-centered Design Thinking methodology to eliminate wasted foot transit and scheduling chaos across VNR VJIET academic corridors.

SK — **Sai Koushik Pallapolu**
Full-Stack Architecture · Real-time SSE Engine & Mobile Responsive Web

T2 — **Co-Developer & Researcher**
User Empathy Research · Corridor Interviews & Student Needs Mapping

T3 — **Systems & Database Lead**
PostgreSQL Schema Design · Prisma ORM & Conflict Scheduling Logic

T4 — **Design & Quality Lead**
UI/UX Design Tokens · Placard Print Generation & Field Verification

OVERVIEW

Agenda & Presentation Flow

Structured through the complete 5-stage Design Thinking journey

Design Thinking
11 Core Milestones

01 Problem Statement
Campus corridor transit friction & zero cabin visibility.

02 Research Approach
Empathy interviews, corridor observations & user mapping.

03 Defining Insights
Categorized user needs: Primary, Secondary, Latent.

04 POV Statement
Actionable Point of View for faculty & students.

05 Ideation Method
Brainstorming & SCAMPER innovation methodology.

06 The Solution
Live QR placards, 1-tap workspace & notify alerts.

07 Our Innovation
Desirability, technical feasibility, and institutional viability.

08 Design for Environment
Sustainable campus infrastructure aligned with UN SDGs.

09 Mentor Interactions
Actionable guidance from Mrs. Prasanna Pabba.

10 Team Experiences
Course Outcomes (CO-1 to CO-4) & member roles.

11 Conclusion & Scope
Deployment roadmap for multi-block campus integration.

Problem Statement

The hidden cost of campus transit and physical availability blindness

CONTEXT & BACKGROUND

The Campus Scale Challenge

Across multi-acre engineering institutions like **VNR VJIET**, faculty members frequently move between lecture halls, laboratory sessions, department evaluation meetings, administrative offices, and research centers.

Students seeking urgent doubt clarification, project reviews, or recommendation signatures routinely walk across Academic Blocks 1, 2, and 3 only to find locked cabins or empty desks.

FIELD OBSERVATION

The Static Placard Breakdown

Traditional door signboards are **completely static**. Even printed paper timetables become inaccurate immediately due to impromptu meetings, lab substitutions, or short corridor discussions.

⚠️ **Core Consequence: Over 40% of student cabin visits end in wasted trips, while faculty experience unplanned drop-in interruptions during grading and research hours.**

01 · DEFINE PHASE

Problem Statement: Core Friction Points

Synthesizing institutional bottlenecks into a clear design challenge

Zero Desk Visibility

Physical presence at desk is unknown until the student physically climbs multiple floors and knocks.

Wasted Time & Energy

Students waste 15–25 minutes between lectures trekking to cabins, causing frustration and crowded corridors.

Scheduling Chaos

Ad-hoc knocking disrupts faculty grading; booking appointments currently requires slow email threads.

No Passive Feedback

When a professor steps away for 10 minutes, students have no way to get alerted when they return.

DESIGN CHALLENGE — HOW MIGHT WE...

"How might we provide students with glanceable, real-time awareness of faculty availability and effortless consultation booking right at the cabin door, while empowering professors with a 1-tap, non-intrusive status management tool?"

02 · EMPATHIZE PHASE

Research Approach & Empathy Map

Field interviews across CSE & IT departments (Cabins 127, 201, 031)

Qualitative Research
Students & Faculty

WHAT USERS SAY

- **Student:** "I climbed to 2nd floor B-Block three times today, but sir was in an evaluation."
- **Professor:** "I step away for 10 minutes to submit a report, and return to find students left."
- **Student:** "I don't know whether to wait outside the door or go back to class."

WHAT USERS THINK

- "Is ma'am actually absent today or just in a quick 5-minute meeting?"
- "If I knock now, am I disturbing an important project review?"
- "I wish there was a live status screen right beside their nameplate."

WHAT USERS DO

- Walk back and forth repeatedly between campus buildings during class breaks.
- Hover awkwardly outside faculty room glass doors trying to peer inside.
- Send repetitive WhatsApp messages or emails asking "Sir are you in cabin?"

WHAT USERS FEEL

- **Frustrated:** Wasting valuable free study periods searching for teachers.
- **Hesitant:** Anxious about knocking on the wrong cabin door during busy hours.
- **Overwhelmed:** Faculty feeling interrupted when focusing on paper evaluation.

Defining Insights: User Needs Matrix

Transforming user empathy into prioritized engineering requirements

NEED TYPE	IDENTIFIED CAMPUS NEED	FIND MY PROFESSOR FEATURE RESPONSE
PRIMARY	**Instant Live Desk Availability:** Students need to verify whether a professor is in office before walking across campus blocks.	**Persistent QR Door Placard & Live Directory:** Instant scan with smartphone camera reveals live status with zero app installation.
PRIMARY	**Expected Return Estimates:** Knowing when a stepped-away professor will be back so students can plan their study schedule.	**Return Time Target Countdown:** Professor sets "+15m", "+30m", or exact return time with optional note (e.g., "In Lab 3").
SECONDARY	**Organized Consultations:** Booking formal 15–30 min doubt slots without back-and-forth emails or double-booking.	**1-Click Consultation Request:** Automatic conflict check preventing overlap; instant professor accept/decline dashboard.
SECONDARY	**Glanceable Office Hours:** Routine availability posted clearly without paper printouts that fade or tear.	**Digital Weekly Office Hours:** Formatted daily schedule with next upcoming availability badge.
LATENT	**Passive Real-Time Awareness:** Relieving students from constantly polling or refreshing their screens.	**"Notify When Available" Subscriber Bell:** Instant push notification / toast alert sent the moment faculty marks "Available".

04 · DEFINE PHASE

Point of View (POV) Statement

Framing the core design challenge through User, Need, and Insight

USER PROFILE

The Academic Community

Undergraduate students seeking guidance and faculty members balancing lectures, research, and consultations across large campus blocks.

SPECIFIC NEED

Zero-Friction Truth

Immediate visibility into real-time physical desk availability and a direct, respectful way to schedule micro-consultations right at the cabin door.

DEEP INSIGHT

Physical-Digital Gap

Physical door signs are completely disconnected from live schedules, while existing ERP apps are too slow and cumbersome for quick 2-minute status updates.

ACTIONABLE POV STATEMENT

"College students seeking academic guidance need a glanceable, instant way to verify faculty availability and request consultations at the door because physical absence is currently invisible until arrived, causing wasted transit time and disruptive interruptions for busy professors."

Ideation Method: Brainstorming & SCAMPER

Systematically exploring alternatives and landing on the optimal solution

Brainstorming Ideas Evaluated

❌ **Biometric/RFID door sensors:** Expensive hardware & privacy concerns.

❌ **Timetable auto-tracking:** Impromptu corridor talks make it inaccurate.

❌ **NFC chips on doors:** Incompatible with older student phones.

✅ **Durable QR Door Placards + Real-time Web App:** Zero hardware cost, universal camera compatibility, sub-second sync.

The SCAMPER Innovation Framework

We applied the SCAMPER creative thinking tool to take traditional door nameplates and radically elevate them into intelligent communication hubs.

S

SUBSTITUTE

Replace static printed door timetables with cloud-linked dynamic QR placards.

C

COMBINE

Combine physical door placarding with automatic real-time push alerts.

A

ADAPT

Adapt real-time flight departure boards for university faculty offices.

M

MODIFY

Modify appointment scheduling into quick 30-min conflict-checked slots.

P

PUT TO USE

Use placards as emergency notices and weekly office hour guidelines.

E

ELIMINATE

Eliminate wasted multi-floor campus trips and disruptive cabin knocking.

R

REARRANGE

Invert booking: students scan and request; faculty approves in 1 tap.

The Solution: Find My Professor

A unified, real-time campus availability & scheduling ecosystem

Full-Stack Solution
Live at VNR VJIET

Core Vision

A frictionless campus bridge that connects physical faculty office doors to students' smartphones in sub-second latency, eliminating wasted foot travel and empowering academic productivity.

Key Differentiator

Zero App Store Friction: Works directly in standard mobile Safari & Chrome via camera scan. Real-time updates push via Server-Sent Events (SSE) with no manual page refreshing needed.

1. Permanent QR Placards

Durable QR codes pasted outside cabin doors. Never need reprinting when timetables change.

2. 1-Tap Faculty Workspace

Professors toggle status (Available, In Class, In Meeting, Stepped Away) with return time in 3 seconds.

3. "Notify When Available" Subscriber Alerts

Students favorite teachers and receive instant live notifications the moment they return to office.

4. Conflict-Checked Micro-Consultations

Students book 30-minute consultation slots; professors accept or decline with 1 click.

Prototype & End-to-End Workflow

Live interactive product demonstration and user touchpoints

Working System
React 19 & Node

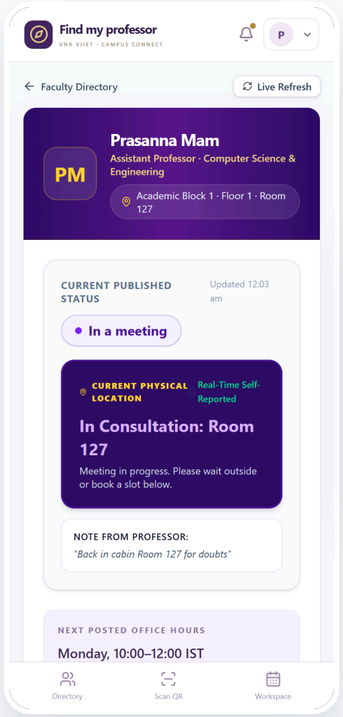
01 Scan Door Placard or Search Directory
Student uses phone camera at cabin door (e.g. Room 127) or browses the campus directory.

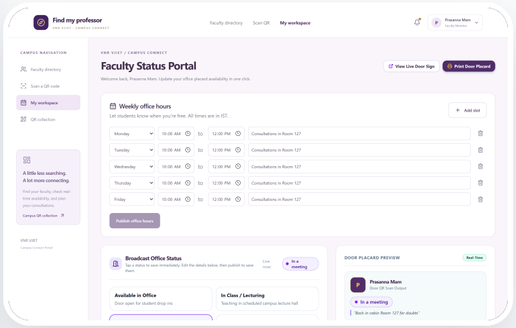
02 View Real-Time Availability & Hours
Live status card renders with color badges, return countdown, and weekly office hours.

03 Tap "Notify When Available"
If teacher is away, student enables the bell toggle to subscribe to live arrival alerts.

04 Request Consultation Slot
Student selects time window and reason; system verifies no overlapping meetings exist.

05 Instant SSE Synchronization
Professor workspace receives instant alert; changes propagate to student screen in sub-second.





Technical Feasibility & Architecture

Engineered for lightweight high-throughput campus execution

Enterprise Stack
PostgreSQL & TypeScript

Technology Stack

Frontend UI: React 19, Vite, Tailwind CSS v4, Lucide Icons

Backend Service: Express.js, TypeScript, Server-Sent Events (SSE)

Database: PostgreSQL, Prisma ORM, JSONB audit logging

QR & Vision: HTML5-QRCode, Canvas 2D Placard Generator

Reactive Real-Time Architecture

Rather than battery-draining client polling, a lightweight SSE stream (`/api/v1/events`) maintains an open connection. On state changes, the server broadcasts an event that prompts connected mobile clients to refetch authorized data in sub-second latency.

Mobile Network Resilience

Mobile browsers frequently sleep SSE connections when screens lock. Our custom `LiveSync` engine listens to `visibilitychange` and network events, automatically re-syncing within 100ms the instant the student turns on their phone.

System Limitations & Mitigation

Limitation: Self-reported availability requires faculty participation.
Mitigation: Configurable TTL (Time-To-Live). If a status expires without confirmation, it automatically displays as "Stale/Unconfirmed" to prevent misleading students.

07 · INNOVATION ANALYSIS

Innovation: Desirability, Feasibility & Viability

Assessing practical sustainability and campus-wide institutional viability

DESIRABILITY

Human Adoption

Students already carry smartphones and instinctively scan QR codes daily. For faculty, a 1-tap mobile toggle respects their autonomy and takes less than 3 seconds—eliminating complex bureaucratic overhead.

FEASIBILITY

Engineering Viability

Built entirely on modern, robust web standards (React 19, TypeScript, PostgreSQL, SSE). Zero hardware dependencies: uses standard laminated QR placards outside faculty rooms with zero maintenance.

VIABILITY

Institutional Value

Can scale across all 15+ engineering departments of VNR VJIET at near-zero incremental cost. Can be hosted on internal college servers or lightweight cloud tiers with negligible bandwidth consumption.

INNOVATION STATEMENT

"Where Is My Professor transforms passive campus doors into interactive, real-time communication portals, bridging the physical-digital divide in higher education through zero-cost QR infrastructure and instantaneous live event streams."

Design for Environment and SDGs

Contributing to institutional sustainability and United Nations Sustainable Development Goals

UN SDGs
3 Core Alignments

04
Quality Education

Target 4.4 & 4.7: Maximizes direct faculty-student mentorship efficiency. Removes scheduling barriers for doubts, final-year project guidance, and research discussions, dramatically enhancing academic productivity.

09
Industry, Innovation & Infrastructure

Target 9.4: Upgrades traditional university infrastructure into a smart, digital campus ecosystem using lightweight web technologies and cloud databases rather than expensive proprietary hardware.

11
Sustainable Cities & Communities

Target 11.6: Reduces wasted inter-building foot transit and elevator electricity consumption across multi-story blocks. Completely eliminates recurring paper reprinting for notices and door schedules.

09 · FEEDBACK & ITERATION

Mentor Interactions & Refinements

Guidance and evolutionary feedback from Mrs. Prasanna Pabba (Assistant Professor, CSE)

STAGE	FOCUS AREA	MENTOR'S GUIDANCE & SUGGESTIONS	ACTION TAKEN BY ENGINEERING TEAM
ITERATION 1	**Initial Problem Definition**	"Do not build an intrusive tracking tool. Faculty will resist any system that attempts automated location surveillance or complex check-in routines."	Pivoted strictly to an **autonomous, self-reported 1-tap workspace** with automated expiration TTLs that respect faculty dignity and privacy.
ITERATION 2	**Physical Door Experience**	"Students who scan the door still need to know general weekly availability when a teacher is stepped away for class."	Integrated **weekly recurring consultation hours** directly into the placard card with automated day-of-week highlighting.
ITERATION 3	**Passive Real-Time Alerts**	"Students waiting outside shouldn't have to keep refreshing their phones. Make the system proactive."	Engineered the **"Notify When Available"** subscriber pipeline and real-time floating toast alerts powered by Server-Sent Events.

10 · REFLECTION & OUTCOMES

Team Experiences and Learnings

Course Outcomes (COs) and collaborative Design Thinking growth

CO-1 · ENGINEERING PROBLEM DEFINITION

Applied the 5 phases of Design Thinking (Empathize, Define, Ideate, Prototype, Test) to solve real-world campus inefficiency.

CO-2 · HUMAN-CENTERED SUSTAINABILITY

Connected user empathy with UN Sustainable Development Goals (SDG 4, 9, 11), saving paper and student transit energy.

CO-3 · RAPID INTERACTIVE PROTOTYPING

Designed, coded, and iterated a production-grade full-stack web application with responsive mobile camera scanning.

CO-4 · REAL-WORLD ENGINEERING DEPLOYMENT

Successfully deployed to cloud infrastructure with managed PostgreSQL, automated migrations, and seed pipelines.

Key Team Takeaway

"The greatest lesson from the CPDT Design Thinking curriculum was that engineering solutions succeed not because of algorithm complexity, but because of how deeply they respect human habits. By eliminating the barrier of installing apps and keeping faculty updates to 1 tap, we turned a perennial campus frustration into a joyful, effortless interaction."

CPDT DESIGN THINKING LAB
VNR Vignana Jyothi Institute of Engineering & Technology

11 · OUTLOOK

Conclusion & Future Scope

Scalable roadmap for institution-wide deployment across VNR VJIET

Project Conclusion

Find My Professor demonstrates how lightweight, modern web software can transform everyday physical architecture into dynamic, connected systems.

The system is fully functional, verified with automated end-to-end tests, supports 23 faculty accounts across 5 engineering branches, and is ready for departmental corridor rollout.

Future Expansion Scope

📍 **Indoor Campus Navigation:** Integrating interactive route maps guiding students from entrance gates directly to Room 127/201.

📟 **Low-Power E-Ink Cabin Displays:** Battery-powered electronic door placards that sync status over Bluetooth/Wi-Fi automatically.

📲 **WhatsApp / Telegram Notification Bot:** Direct push alerts to students' messaging apps when favorited faculty returns.

🔗 **Official ERP (Automation4U) Integration:** Auto-populating verified class schedules and semester examination duties.

"We don't replace personal faculty-student mentorship; we remove the friction that stands in its way."

VNR VJIET · CPDT 2026

Thank You!

Open for Questions, Demonstration & Feedback

PROJECT MENTOR

Mrs. Prasanna Pabba

Assistant Professor, Department of Computer Science & Engineering · VNR VJIET